

Generation Interconnection System Impact Study Report

For Queue Project Q18-1008

Firewheel 138 kV

October 21, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1008. The Interconnection Customer proposes a 29.6 MW solar photovoltaic facility interconnecting at the Firewheel 138 kV substation in the West Virginia area. The request was received on June 04, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 368 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,669,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Abbotsford-Bridge River 138kV L
Contingency: American-Urban 345kV L
Loading: 119.8% (Rating: 222 MVA)
DFAX: 11.9%
Pre-project loading: 118.2%
2. Facility: Artesia Country Club 345-1 kV XFMR
Contingency: Air Force Base-Artesia Country Club 345kV
Loading: 110.8% (Rating: 540 MVA)
DFAX: 7.6%
Pre-project loading: 110.4%
3. Facility: Brooklyn City N Plant-Midset 138kV
Contingency: American-Urban 345kV
Loading: 107.7% (Rating: 277 MVA)
DFAX: 21.6%
Pre-project loading: 105.4%
4. Facility: Rochester Hydro-Depere 138kV L
Contingency: Clark County-Kearneyville 345kV
Loading: 107.6% (Rating: 212 MVA)
DFAX: 6.5%
Pre-project loading: 106.7%
5. Facility: Artesia Country Club-Ross Fork Substation 138kV Line
Contingency: Artesia Country Club-Air Force Base 345kV line
Loading: 105.3% (Rating: 647 MVA)
DFAX: 7.7%
Pre-project loading: 105.0%
6. Facility: Artesia Country Club 138/1kV Transformer Bank
Contingency: Artesia Country Club-Air Force Base 345kV
Loading: 104.1% (Rating: 575 MVA)

DFAX: 7.6%

Pre-project loading: 103.7%

7. Facility: Stoneall-Redbud 138kV line

Contingency: Clark County-Kearneyville 345kV

Loading: 103.6% (Rating: 264 MVA)

DFAX: 5.4%

Pre-project loading: 103.0%

8. Facility: Santa Rosa Energy Center-Balch 138kV line

Contingency: Clark County-Kearneyville 345kV

Loading: 100.5% (Rating: 318 MVA)

DFAX: 5.4%

Pre-project loading: 100.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Abbotsford-Bridge River 138kV L	Rebuild line with higher-capacity conductor	\$7,920,000	100% (\$7,920,000)
Artesia Country Club 345-1 kV TX	Replace transformer with higher-rated unit	\$21,061,000	12% (\$2,519,000)
Midset-Brooklyn City N Plant 138kV	Reconductor line	\$2,774,000	100% (\$2,774,000)
Rochester Hydro-Depere 138kV L	Reconductor line	\$10,252,000	3% (\$343,000)
Artesia Country Club-Ross Fork Substation 138kV Line	Reconductor line	\$1,052,000	8% (\$86,000)
Artesia Country Club 138/1kV Transformer Bank	Replace transformer with higher-rated unit	\$5,023,000	12% (\$601,000)
Stonewall-Redbud 138kV	Reconductor line	\$1,763,000	13% (\$236,000)
Balch-Santa Rosa Energy Center 138kV	Reconductor line	\$1,421,000	13% (\$190,000)
	Total allocated to project		\$14,669,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.